Quotient Space

Def. Let $(X, \mathcal{L})$ be a Topological space, and Y be a set.
For an onto map $f: X \to Y$, define a Quotient Topology on Y induced by f as

$$\mathcal{L}_Q \overset{\text{def}}{=} \{U \subseteq Y \mid f^{-1}[U] \text{ is open of } (X, \mathcal{L})\}.$$

It is actually a Topology on Y, because: clearly, $\emptyset, Y \in \mathcal{L}_Q$ and

$U_\alpha \in \mathcal{L}_Q \ (\alpha \in \Lambda) \Rightarrow f^{-1}\left[\bigcup_{\alpha\in\Lambda} U_\alpha\right] = \bigcup_{\alpha\in\Lambda} f^{-1}[U_\alpha]$ is also open in X, so is $\bigcup U_\alpha$.

Similarly, $f^{-1}\left[\bigcap_{\alpha\in\Lambda} U_\alpha\right] = \bigcap_{\alpha\in\Lambda} f^{-1}[U_\alpha]$ also open if Λ is finite, so is $\bigcap U_\alpha$.

Moreover, it is the finest Topology on Y such that f is Continuous, because
$f: X \to Y$ is continuous iff $\forall U \in \mathcal{L}_Y$, $f^{-1}[U] \in \mathcal{L}_X$ iff $\mathcal{L}_Y \subseteq \mathcal{L}_Q$.

where $\mathcal{L}_Y$ is arbitrary Topology on Y. □

Def. Let X be a Topological Space, and $\sim$ be an equivalence relation on X.
Define the Canonical map on X as

$$\pi: X \to X/\sim;\ x \mapsto [x]$$

The $(X/\sim, \mathcal{L}_\pi)$ is called Quotient Space where $\mathcal{L}_\pi$ is a Quotient topology on $X/\sim$, induced by π. That is,

$$\mathcal{L}_\sim = \{U \subseteq X/\sim \mid \pi^{-1}[U] \text{ is open}\},$$

Note: For any $U \subseteq X/\sim$, $\pi^{-1}[U] = \{x \in X \mid \pi(x) \in U\} = \bigcup_{[x]\in U} [x]$
and for any $A \subseteq X$, $\pi^{-1}[\pi[A]] \overset{\text{def}}{=} \bigcup_{x\in A} [x]$.

$\text{Sat}_f(A)$, f-Saturation of A.

[Saturation.

[Prop. Let X be a Topological Space, and let $\sim$ be equivalent relation.
For $A \subset X$, $\pi[A] \subseteq X/\sim$ is open $\Leftrightarrow$ $Sat(A) \subset X$ is open.

where $\pi : X \to X/\sim : x \mapsto [x]$ is a Canonical Quotient map,
and $Sat(A) = \pi^{-1}[\pi[A]] = \bigcup_{x \in A} [x]$.

Note: Generally, $A \subset Sat(A)$.

Proof. By definition of Quotient Topology,
$\pi[A]$ is open $\Leftrightarrow$ $\pi^{-1}[\pi[A]]$ is open. ☑

Let X be a Topological Space, $\sim$ be an equivalent relation on X, π be a Canonical map.

Lemma. Let Z be a Topological Space and a map $g: X/_\sim \to Z$,

g is Continuous if and only if $g \circ \pi : X \to Z$ is Continuous.

Pf. Since $\pi : X \to X/_\sim$ is Continuous, trivially $g \circ \pi$ is Continuous.

Conversely, assume $g \circ \pi$ is Continuous. Then, given an open $U \subseteq Z$,

$(g \circ \pi)^{-1}[U] = \pi^{-1}[g^{-1}[U]]$ is open of X, so $g^{-1}[U]$ is open of $X/_\sim$. ∎

Lemma. Let Z be a Topological Space.

If a Continuous map $f: X \to Z$ satisfies $x \sim y \Rightarrow f(x) = f(y)$,

then a map induced by f:

$$\overline{f} : X/_\sim \to Z : [x] \mapsto f(x)$$

is Continuous and it is a unique map such that $\overline{f} \circ \pi = f$.

Pf. $\overline{f}$ is well-defined by: $[x] = [y] \Leftrightarrow x \sim y \Rightarrow f(x) = f(y)$.

Given $x \in X$, $f(x) = \overline{f}([x]) = \overline{f}(\pi(x)) = (\overline{f} \circ \pi)(x)$, so $\overline{f} \circ \pi = f$

By the Lemma above, $\overline{f}$ is Continuous, And, given a map $g: X/_\sim \to Z$ s.t. $g \circ \pi = f$,

$$g([x]) = g(\pi(x)) = f(x) = \overline{f}([x])$$

So the uniqueness is guaranteed.

Lemma. Let Z be a Topological Space.

If a Continuous onto map $f: X \to Z$ satisfies $x \sim y \Leftrightarrow f(x) = f(y)$ and either open or closed,

then a map induced by f:

$$\overline{f} : X/_\sim \to Z : [x] \mapsto f(x)$$

is Homeomorphism.

[Lemma. If a continuous onto open (or closed) map $f: X \to Y$ satisfies

$$x \sim y \iff f(x) = f(y)$$

Then the induced map

$$\tilde{f}: X/\sim \to Z : [x] \mapsto f(x)$$

is Homeomorphism.

Proof. Notice that $f = \tilde{f} \circ \pi$. Suppose that f is an open map. Then, given open set $U \subset X/\sim$, $\pi^{-1}[U]$ is open. And, $U = \pi[\pi^{-1}[U]]$,

$$\tilde{f}[U] = \tilde{f}[\pi[\pi^{-1}[U]]] = f[\pi^{-1}[U]] \subset Z$$ is open, since f is open map

Similarly, given closed set $C \subset X/\sim$,

$$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$$ is closed if f is closed map. □

[Quotient map.

[Def. Given Topological space X and Y, a map $f: X \to Y$ is called a quotient map if f is Continuous, Onto, and satisfies for any $U \subseteq Y$,

$$U \text{ is open in } Y \overset{f \text{ cont}}{\underset{\text{Quotient}}{\Leftrightarrow}} f^{-1}[U] \text{ is open in } X.$$

[Prop. Composition of Quotient map is also Quotient map.

Proof Let $f: X \to Y$ and $g: Y \to Z$ be Quotient maps. Given open set $U \subseteq Z$,

$(g \circ f)^{-1}[U]$ is open $\Leftrightarrow f^{-1}[g^{-1}[U]]$ is open $\Leftrightarrow g^{-1}[U]$ is open $\Leftrightarrow U$ is open

And, $(g \circ f)[X] = g[f[X]] = g[Y] = Z$, $(g \circ f)$ is a Quotient map. ∎

[Prop. If a Continuous Onto map $f: X \to Y$ is either open or closed map, then f is Quotient map.

Proof Given $U \subseteq Y$, if $f^{-1}[U]$ is open then $f[f^{-1}[U]] \overset{\text{onto}}{=} U$ is open, if f is open map.

And, if $f^{-1}[U]$ is open, then $f[X \setminus f^{-1}[U]]$ is closed if f is closed map.

Since f is onto, $X \setminus f^{-1}[U] = f^{-1}[Y] \setminus f^{-1}[U] = f^{-1}[Y \setminus U]$, so

$$f[X \setminus f^{-1}[U]] = f[f^{-1}[Y \setminus U]] = Y \setminus U \text{ is closed, or } U \text{ is open.}$$ ∎

But, there is a quotient map neither open nor closed: define $X = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0 \text{ or } y = 0\}$.

$$f: X \to \mathbb{R};\ (x,y) \mapsto x$$

Then, it is clearly onto, and Continuous is given by: for any $a < b$ in $\mathbb{R}$,

$$f^{-1}[(a,b)] = [(a,b) \times \mathbb{R}] \cap X \text{ is open.}$$

To show f is quotient map, suppose that $f^{-1}[U]$ is open for some $U \subseteq \mathbb{R}$.

Given $x \in U$, $f((x,0)) = x \in U$, so $(x,0) \in f^{-1}[U]$. Since $f^{-1}[U]$ is open, for some $\varepsilon > 0$,

$(x,0) \in B_\varepsilon((x,0)) \cap X \subseteq f^{-1}[U]$. Thus, $f[B_\varepsilon((x,0))] = (x-\varepsilon, x+\varepsilon) \cap f[X] = (x-\varepsilon, x+\varepsilon) \subseteq U$.

So, U is open.

Moreover, $B_1((0,2)) \cap X$ is open of X, but $f[B_1((0,2)) \cap X] = [0,1)$, not open.

And, $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$ is closed because $F' = \emptyset$, so $F = \overline{F}$.

But, $f[F] = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ is not closed in $\mathbb{R}$, since $0 \notin f[F]$ but $0 \in f[F]'$.

Comment: A projection $\pi: \mathbb{R}^2 \to \mathbb{R}$ cannot be Closed map: $F = \{(\frac{1}{n}, n) \mid n \in \mathbb{N}\}$.

Quotient map.

Def. Let X, Y be Topological spaces

A onto map $f: X \to Y$ satisfying

$$U \subseteq Y \text{ is open} \iff f^{-1}[U] \subseteq X \text{ is open.}$$

Note: A space X and a onto map $f: X \to Y$, let Y be equipped:

$$\mathcal{U}_Q := \{U \subseteq Y \mid f^{-1}[U] \subseteq X \text{ is open}\}.$$

Then, f is a quotient map.

Lemma. Let $f: X \to Y$ be a Continuous onto map.

Then, f is a quotient map if and only if

given a space Z and a map $g: Y \to Z$, g is conti. $\iff g \circ f$ is conti.

[Prop. Let X be a Topological space and $\sim$ be an equivalent relation on X
If the Canonical Projection $\pi: X \to X/\sim : x \mapsto [x]$ is open map,
then for a basis $\mathcal{B}$ for X, $\mathcal{B}_\sim = \{\pi[B] \mid B \in \mathcal{B}\}$ is a basis for $X/\sim$.

Proof. Let $\mathcal{U} \subseteq X/\sim$ be an open set, and let $[p] \in \mathcal{U}$ be given.
Since $\mathcal{U}$ is open, $\pi^{-1}[\mathcal{U}] \subseteq X$ is open in X.
Meanwhile, for $p \in [p]$ $\pi(p) = [p] \in \mathcal{U}$, so $p \in \pi^{-1}[\mathcal{U}]$. Thus, there is $B \in \mathcal{B}$ such that

$$p \in B \subseteq \pi^{-1}[\mathcal{U}].$$

Now, since π is onto,

$$[p] = \pi(p) \in \pi[B] \subseteq \pi[\pi^{-1}[\mathcal{U}]] = \mathcal{U}$$

Since π is open map, $\pi[B]$ is open, so $\mathcal{B}_\sim$ is a basis for $X/\sim$. ∎

Note: so, if π is open, then X 2nd-countable $\Rightarrow X/\sim$ ".

Examples.

Define equivalence relation $\sim$ on $\mathbb{R}$ generated by $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$.

Then, $\mathbb{R}/\sim = \mathbb{R}/\mathbb{Z} = \{a + \mathbb{Z} \mid a \in \mathbb{R}\} = \{a + \mathbb{Z} \mid a \in [0,1)\}$.

$\mathcal{U} \subseteq \mathbb{R}/\sim$ is open set if and only if $\pi^{-1}[\mathcal{U}]$ is open,

where $\pi : \mathbb{R} \to \mathbb{R}/\sim ; x \mapsto x + \mathbb{Z}$.

So, the topology $\mathcal{O}_\sim$ on $\mathbb{R}/\sim$ is generated by $\mathcal{B}_\sim = \{(a,b) + \mathbb{Z} \mid a < b\}$, because:

Given $(a,b) \subseteq \mathbb{R}$, $\pi[(a,b)] = (a,b) + \mathbb{Z} = \{r + \mathbb{Z} \mid a < r < b\}$,

and $Sat_\pi(a,b) = \pi^{-1}[\pi[(a,b)]] = \bigcup_{r \in (a,b)} r + \mathbb{Z} = \bigcup_{n \in \mathbb{Z}} (a+n, b+n)$.

Since the saturation is open, so $\pi[(a,b)]$ is also open. Therefore,

the Canonical projection π is open map, so the $\mathcal{B}_\sim$ is basis for $\mathbb{R}/\sim$.

Moreover, $\mathbb{R}/\sim$ is Homeomorphic to S^1, because i define a map

$$f : \mathbb{R} \to S^1 ; x \mapsto e^{2\pi i x}.$$

Then, f is continuous onto and open map satisfying

$$x \sim y \Leftrightarrow x - y \in \mathbb{Z} \Leftrightarrow e^{2\pi i (x-y)} = 1 \Leftrightarrow f(x) = f(y)$$

So, the induced map

$$\tilde{f} : \mathbb{R}/\sim \to S^1 ; [x] \mapsto f(x)$$

is well-defined and injective, and since f and π are onto and $\tilde{f} \circ \pi = f$,

$$\tilde{f}[X/\sim] = \tilde{f}[\pi[X]] = (\tilde{f} \circ \pi)[X] = f[X] = Y.$$

So $\tilde{f}$ is also surjection. And, given open $\mathcal{U} \subseteq S^1$, $f^{-1}[\mathcal{U}]$ is open and

$\tilde{f}^{-1}[\mathcal{U}]$ is open $\Leftrightarrow \pi^{-1}[\tilde{f}^{-1}[\mathcal{U}]]$ is open $\Leftrightarrow f^{-1}[\mathcal{U}]$ is open

Therefore $\tilde{f}$ is also continuous. Finally, since the f is open map, given open $\mathcal{V} \subseteq \mathbb{R}/\sim$,

$\tilde{f}[\mathcal{V}] = \tilde{f}[\pi[\pi^{-1}[\mathcal{V}]]] = f[\pi^{-1}[\mathcal{V}]]$ is open, so

Continuous open bijective map $\tilde{f}$ is Homeomorphism. □